Paleosecular Variation of the Honolulu Volcanic Series, Oahu, Hawaii

RICHARD R. DOELL

U.S. Geological Survey, Menlo Park, California 94025

A paleomagnetic study of 193 sample cores drilled from 25 sites (lava flows) in the Honolulu volcanic series exposed on SE Oahu, Hawaii, yielded a mean angular standard deviation of 9.4°, with 95% confidence limits between about 7.7° and 11.9°. Thus, these lavas in the Brunhes normal polarity epoch confirm an earlier low value for paleosecular variation of 11.0° in the central Pacific determined on lavas of similar age from the Island of Hawaii.

The absence at present of nondipole geomagnetic field components in a large area of the central Pacific Ocean [*Fisk*, 1931; *Cox*, 1962; *Creer*, 1962; *Doell and Cox*, 1972] is a very unusual feature of the earth's magnetic field. The question of the permanence of this feature has been pursued in several paleomagnetic investigations [*Doell and Cox*, 1963, 1965; *Doell*, 1969] and discussed in two recent contributions [*Doell and Cox*, 1971, 1972]. The present study of 27 lava flows in the Honolulu volcanic series is a further contribution to this question.

Details of the age relations of lava flows in the Honolulu volcanic series, like those of other Brunhes age lava flows from the Island of Hawaii [*Doell and Cox*, 1963, 1965; *Doell*, 1969], are not well known. For this series, however, there is good geological and radiometric-age information to suggest that a relatively long period of time was involved in the eruption. This information does not imply that the several volcanic series on Hawaii, which consist of thick sections of superposed lava flows, were all erupted in very short-time periods. The evidence suggesting longer time periods has been discussed in the previous studies. The areally scattered eruptions of the Honolulu volcanic series have been related to ancient sea stands, which provide a more definite estimate (as do the radiometric-age data) of the total period of time required for their emplacement. In any paleosecular variation study designed to study nondipole field changes, it is important that an interval of time sufficient to cover all periods represented in geomagnetic secular variation due to the nondipole field be sampled. The maximum periods are estimated to be about 1000 years [*Cox and Doell*, 1964]. Like Brunhes age Hawaiian lava flows of earlier studies, the Honolulu volcanic series yields a very low value for paleosecular variation; this low value indicates a subdued or absent nondipole field in this region during the Brunhes normal polarity epoch.

Copyright © 1972 by the American Geophysical Union.

GEOLOGICAL SETTING

The Honolulu volcanic series consists of about 30–40 areally scattered lava flows and pyroclastic deposits that were erupted over SE Oahu after a rather long period of erosion following the construction of the Koolau Range. The occurrence and geologic relations of these rocks are discussed in detail by *Stearns and Vaksvik* [1935] and *Winchell* [1947]. The generalized map shown in Figure 1 is taken from *Winchell* [1947].

Several terraces and platforms above and below the present sea level have been recognized on Oahu and interpreted as ancient shorelines [*Stearns*, 1935; *Ruhe et al.*, 1965]. *Stearns* [1966] has divided the Honolulu volcanic series into four age groups on the basis of their relations to the ancient sea stands [*Stearns*, 1966, pp. 96–97]. In brief, the oldest lavas sampled for the present study, numbers HS 1, 2, and 3 (Figure 1 and Table 1), were erupted during the Kahipa (−92 ± meter) stand; the succeeding lavas, numbered HS 4 through 14, were formed during the Kaena (+29 meter) and Laie (+22 meter) stands; the third group, numbers HS 15 through 22, were erupted during the

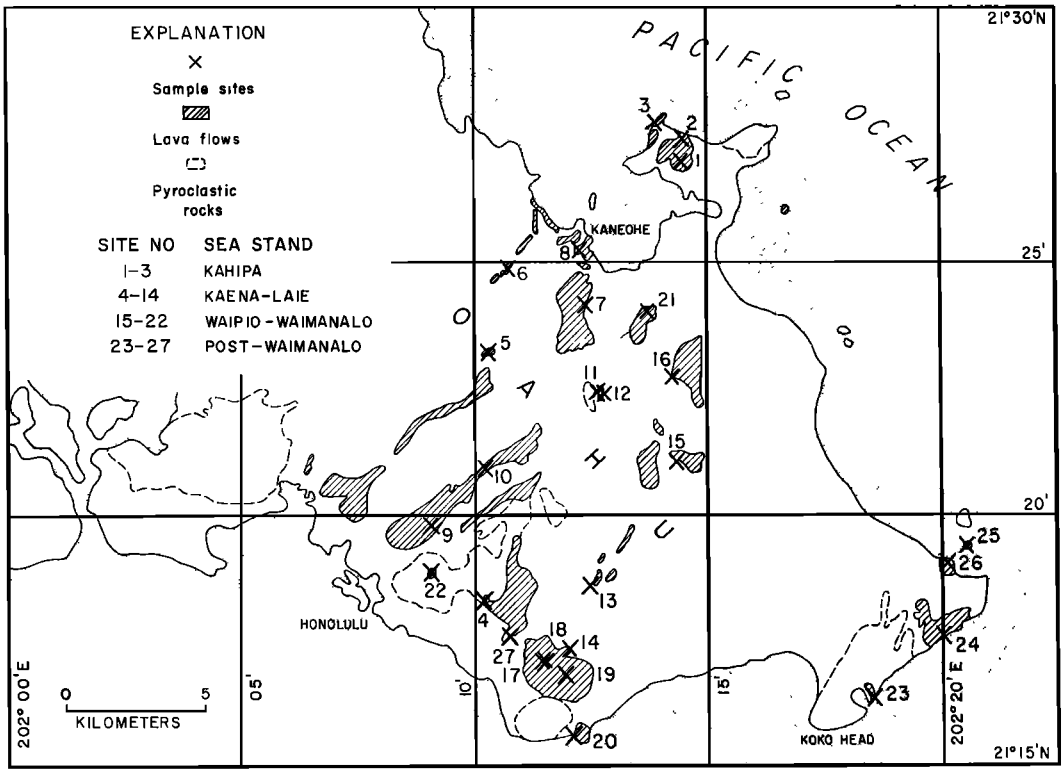


Fig. 1. Generalized geologic map of the Honolulu volcanic series in southeastern Oahu [after *Winchell*, 1947] showing location of sampling sites. Also indicated are age relations of the lavas to ancient sea stands.

Waipio (−18 meter) and Waimanalo (+8 meter) stands; and a final group, numbers HS 23 through 27, are post-Waimanalo stand. *Ruhe et al.* [1965] question the reality of a Laie stand, but they find much evidence for the pronounced Kaena and Waimanalo stands, which are of major importance in the relative dating of the Honolulu volcanic series.

The various sea stands on Oahu have been related to glacio-eustatic changes of sea level. On the basis of several radiocarbon dates, *Ruhe et al.* [1965] assign a late Sangamon age to the Waimanalo stand. (The Waimanalo stand grades seaward to a −92-meter submarine shelf correlated with the Wisconsin maximum.) This assignment would suggest that lavas numbered HS 15 through HS 27 are younger than about 70,000 years. The Kaena stand has been related to the Yarmouth interglaciation [*Ruhe et al.*, 1965, Table 1] of earlier but relatively uncertain age. *Broecker et al.* [1968] have obtained radiometric ages related to the last few paleo sea-level stands, going back some 120,000 years.

Preliminary K/Ar determinations on two lavas related to the Kaena stand (HS 9 and HS 14) yield ages near 350,000 B.P. (G. B. Dalrymple, personal communication, 1970), which is not in disagreement with the correlation between the Kaena stand and the Yarmouth interglaciation.

Gramlich et al. [1971] have recently obtained K/Ar ages on 12 lavas of the Honolulu volcanic series with values of the order of 30,000, 70,000, 290,000, 430,000, 650,000, and 850,000 years represented. Although high precision in individual determinations is difficult to obtain in rocks of such young age, there is a strong indication from these data that eruption of the Honolulu volcanic series took place throughout the Brunhes polarity epoch.

There is some evidence, such as marked differences in weathering, for different times of

21562202, 1972, 11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB077i011p02129 by Curtin University Library, Wiley Online Library on [26/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. Flow-Average Data

Unit No.	No. of Samples*	af Cleaning (Peak), oe	RM Declination, °E	RM Inclination, deg Dn	Length of Resultant R	Precision Parameter k_e	95% Confidence, deg	Angular s.d., deg	VGP Lat., °N	VGP Long., °E
HS 27	8(8)	200	9.6	35.2	7.9716	247	3.5	5.2	80.8	301.9
HS 26	7(8)	200	12.7	36.9	6.9440	107	5.9	7.8	78.1	294.0
HS 25	7(8)	400	7.8	41.5	6.9319	88	6.5	8.6	82.4	271.6
HS 24	7(8)	200	8.3	33.3	6.9816	326	3.4	4.5	81.6	312.9
HS 23	8(8)	NRM	2.4	36.4	7.9765	298	3.2	4.7	87.5	316.8
HS 22	8(8)	NRM	9.3	35.1	7.9896	671	2.2	3.1	81.1	303.0
HS 21	7(8)	100	352.8	23.3	6.8939	57	8.1	10.8	78.4	60.0
HS 20	7(8)	NRM	6.7	46.9	6.9918	732	2.2	3.0	80.9	242.4
HS 19	8(8)	NRM	2.7	37.8	7.9532	150	4.6	6.6	87.5	293.2
HS 18	8(8)	100	357.0	36.2	7.9702	235	3.6	5.3	86.9	89.6
HS 17	8(8)	200	356.3	38.8	7.9635	192	4.0	5.9	86.5	123.2
HS 16	8(8)	100	336.9	26.7	7.9607	178	4.2	6.1	66.9	97.8
HS 15	7(8)	100	348.8	27.8	6.9603	151	4.9	6.6	77.4	82.4
HS 14	8(8)	400	353.6	39.2	7.9578	166	4.3	6.3	84.0	121.6
HS 13	8(8)	25	355.9	35.0	7.9262	95	5.7	8.3	85.7	85.0
HS 12	7(7)	NRM	354.7	29.7	6.9787	281	3.6	4.8	82.6	65.6
HS 11	8(8)	NRM	352.1	34.4	7.9914	811	2.0	2.8	82.2	95.0
HS 10	8(8)	NRM	358.9	20.0	7.9354	108	5.3	7.8	78.9	27.8
HS 9	8(8)	200	3.0	25.2	7.8806	59	7.3	10.6	81.5	2.5
HS 8	8(8)	50	2.8	22.3	7.9781	319	3.1	4.5	79.8	6.5
HS 7	8(8)	NRM	4.3	37.3	7.9636	192	4.0	5.8	85.9	299.0
HS 6	8(8)	NRM	0.9	27.8	7.9594	172	4.2	6.2	83.3	14.9
HS 5	8(8)	50	359.4	8.2	7.9854	479	2.5	3.7	72.7	24.3
HS 4	8(8)	50	10.2	41.6	7.9804	357	2.9	4.3	80.2	274.8
HS 3†	8(8)	400	317.3	49.5	6.5836	5	27.8	37.3	50.8	134.4
HS 2	8(8)	100	357.0	19.1	7.9453	128	4.9	7.2	78.0	36.8
HS 1†§										

*Number of samples collected; number in parentheses is number of samples used in flow-average analysis.
†Not included in analyses for paleosecular variation.
§The af demagnetization of all cores shows strong anomalous RM.

21562202, 1972, 11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB077i01p02129 by Curtin University Library, Wiley Online Library on [26/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

eruption within the given groups, and *Stearns* [1966, p. 100] suggests that up to the present time Honolulu volcanic series volcanism has been more or less continuous throughout the history of changing sea level. This conclusion is also supported by the more recent K/Ar data. The Honolulu volcanic series should therefore be a suitable series of lavas for random sampling of the earth's magnetic field in the central Pacific during the last, or Brunhes, geomagnetic polarity epoch.

METHODS AND RESULTS

Not all the eruptions that make up the Honolulu volcanic series produced lava flows suitable for paleomagnetic studies; nor was it possible to locate and sample every unit recognized and described as a lava flow by *Stearns and Vaksvik* [1935] and *Winchell* [1947]. It was possible to sample 27 sites, as shown in Figure 1. Specific details of the locations are given in the appendix.

All samples were drilled and geographically oriented in situ by using portable equipment, remanent magnetization (RM) was measured with spinner type magnetometers, and samples were tested for remanence stability and magnetically cleaned by using partial alternating field (af) demagnetization.

In general, one specimen from each of eight cores drilled at a given site was measured; flow-average paleomagnetic data for each site are given in Table 1. Various amounts of af cleaning were needed to remove anomalous magnetization (column 3 of Table 1), and in a few cases specimens with widely divergent directions were excluded from analysis (column 2 of Table 1). The general rationale used in selecting a particular af field for magnetic cleaning and for excluding specimens from flow-average analysis is considered in detail in an earlier paper [*Doell*, 1970]. For two sites (HS 1 and 3) it was not possible to recognize clearly the direction of the original thermoremanent magnetization.

The over-all stability of the natural remanent magnetization (NRM) of the Honolulu volcanic series is summarized in Figure 2 in the form of a series of histograms of the intensity of RM remaining after partial af demagnetization treatments at progressively higher peak field strengths. The measurements are on one core per lava flow. The RM typically decreases from a value of about 5×10^{-3} emu/cc before treatment to about one-tenth this intensity after af treatment at 800-oe peak field strength. This behavior typifies young Hawaiian lavas [*Doell and Cox*, 1963, 1965; *Doell*, 1969] and suggests that the NRM is predominantly TRM acquired when the lavas cooled. Thus the declination and inclination data cited in Table 1 can be taken as those of the geomagnetic field at the time and place of cooling of the individual lavas. The precision with which these directions have been determined is given by the values of the precision parameter k, the 95% confidence semiangle α_{95} [*Fisher*, 1953], and the angular standard devia-

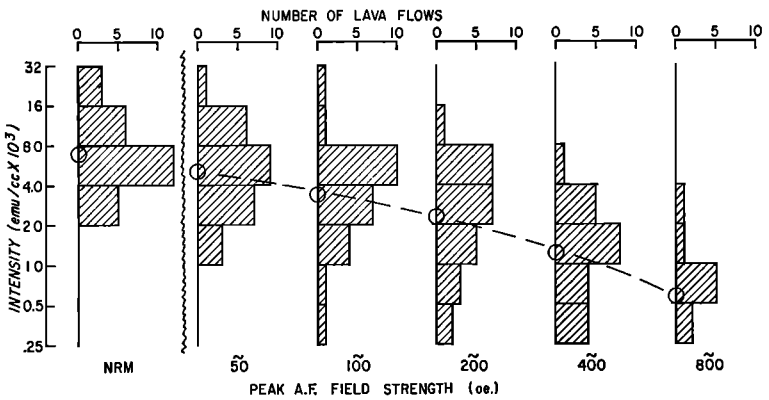


Fig. 2. Histograms of remanent magnetization remaining after partial alternating-field demagnetization at peak field strengths of 0 (NRM), 50, 100, 200, 400, and 800 oe.

21562202, 1972, 11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB077i011p02229 by Curtin University Library, Wiley Online Library on [26/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

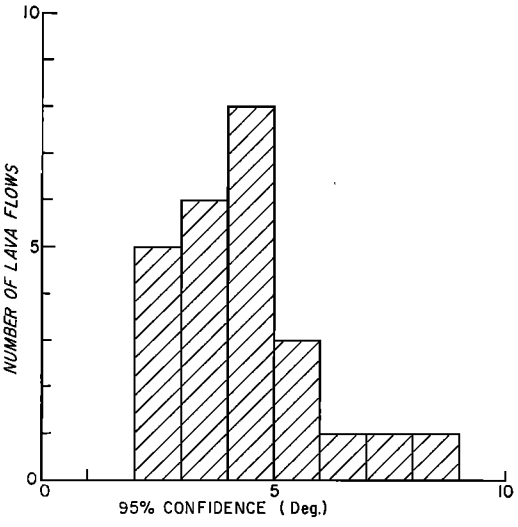


Fig. 3. Histogram of flow-average 95% confidence cone semiangles (α_{95}).

tion. The over-all precision is summarized in Figure 3 as a histogram of the α_{95} values, where it is seen that the majority of the directions are defined within some 5° to 6°.

DISCUSSION

The positions of the virtual geomagnetic poles (VGP) calculated from the directions of the cleaned RM cited in Table 1 are shown in Figure 4, along with the mean position and its 95% confidence circle. The means for both VGP's and RM directions and certain measures of variance used to define paleosecular variation are given in Table 2. The methods used in calculating these are based on procedures developed by *Fisher* [1953], *Watson and Irving* [1957], and *Cox* [1970]. The exact manner in which these procedures were employed is given in an earlier publication [*Doell*, 1970]. A value of $S_A = 1.25°$ was assumed as the contribution to the total variance due to local magnetic anomalies at the time of the cooling of the lavas [*Doell and Cox*, 1963] and was used in the conversion from between-flow variance (subscript B, Table 2) to ancient geomagnetic variance (subscript F).

In calculating the 95% confidence limit for the over-all mean and in calculating confidence limits for the measures of variance, it was assumed that each lava flow represented a separate sample of the geomagnetic field drawn at random with respect to time. Naturally, if two or more lava flows sampled were really part of a single volcanic episode that lasted less than about a few years, this assumption would not be valid, and the effective number of lava flows sampled would be reduced.

In the present study, there are several instances in which the lava flows may be part of a single volcanic episode. For example, the four lava flows sampled from the Koko fissure (HS 26 through 23) may be the same age if this fissure was active for only one volcanic episode. Other sites that may yield redundant data, based on geological reasoning, are HS 19-18-17, HS 12-11, HS 10-9, and HS 8-7. One way to test for such redundancies is to use *Watson's* [1956] test for the 'comparison of two polar vectors.' This test, which determines whether two populations of directions (in this case the directions measured in individual cores from two lavas) are the same, is permissive but not conclusive evidence that the lavas belong to a single volcanic episode, because the geomagnetic field reoccupies the same direction many different times. On the other hand, if the two mean directions differ significantly, then the lavas must have been erupted at different times. The test was made on pairs of lavas from the groups mentioned, by using a confidence level of 95%. Of the four Koko fissure eruptions, HS 26 could be the same as either HS 25 or 24, but HS 25, 24, and 23 are all different from each other. HS 8 and 7 also are definitely different from each other, but the lavas within the other three groups may be from only three different eruption episodes. Thus, in the Honolulu volcanic series it is possible that the effective number of lavas sampled should be reduced from 25 to 20. The truth probably lies somewhere between these two values.

The mean VGP for the Honolulu volcanic series paleomagnetic data lies almost directly on the far side of the geographic pole from the sampling site. *Wilson* [1970] has noted this same feature for paleomagnetic results from many parts of the world. He has suggested that this fact might indicate a permanent displacement of the 'main' axial dipole toward the north pole. At the 95% confidence level the mean VGP position for the Honolulu volcanic series is sig-

21562202, 1972, 11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB077i011p02129 by Curtin University Library, Wiley Online Library on [26/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

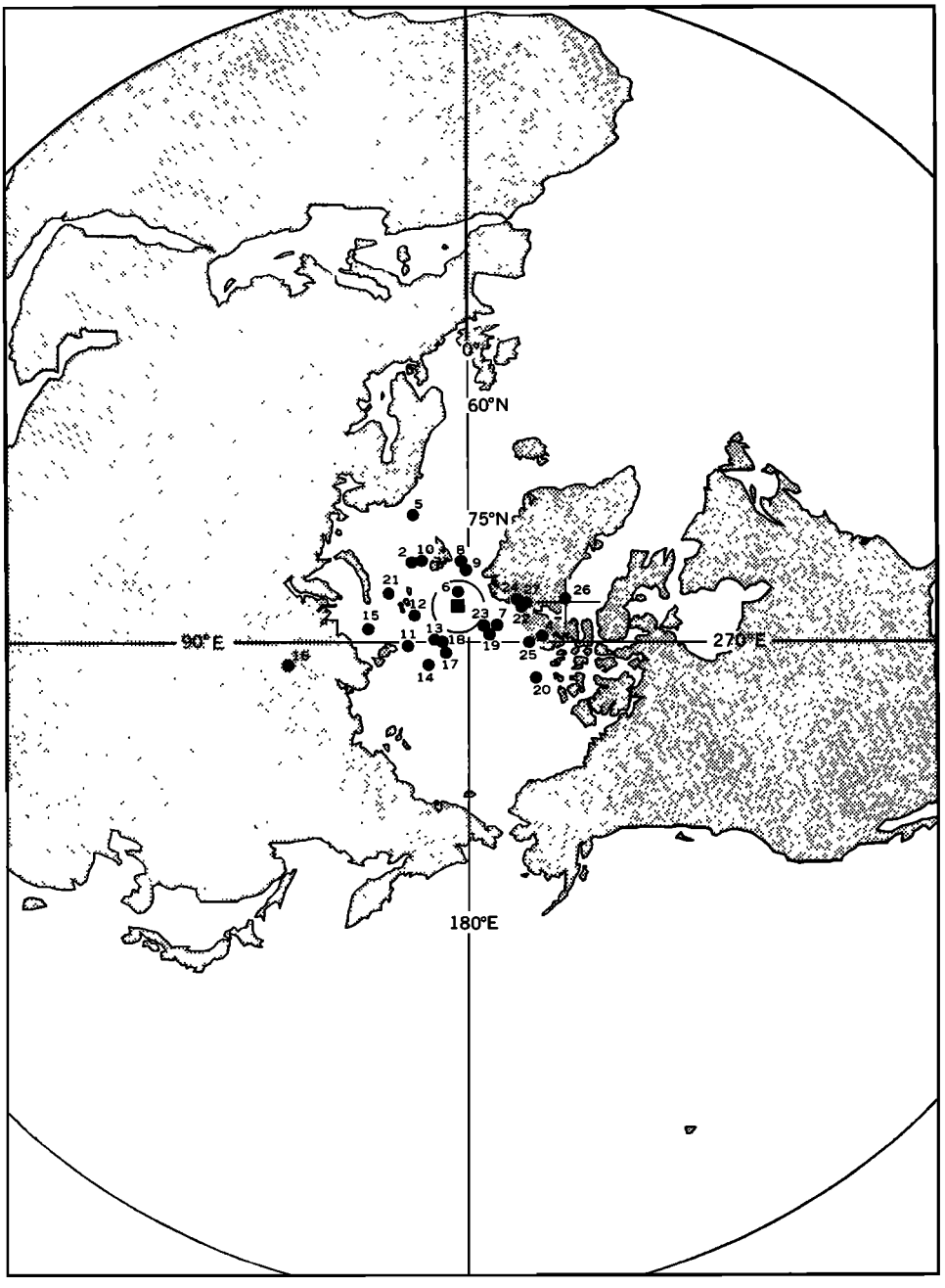


Fig. 4. Virtual geomagnetic pole positions of lavas in the Honolulu volcanic series. Numbers refer to those of column 1, Table 1. Square is the over-all mean, with the 95% confidence circle drawn around it ($N = 25$). Cross is the VGP determined from the present field direction in Hawaii. Portion of Lambert's equal-area projection.

21562202, 1972, 11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB077i011p02129 by Curtin University Library, Wiley Online Library on [26/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2. Statistical Analyses of Flow-Average Data

Virtual Geomagnetic Poles				RM Directions			
Lat.	Long.	R	α_{95}	Inclin.	Declin.	R	α_{95}
86.4°	20.9°	24.679	3.4°	32.1°	359.9°	24.556	4.0°
K_T	K_W	K_B	K_F	k_T	k_W	k_B	k_F
75	72	88	90	54	163	57	57
67		*77*	*78*	*43*		*45*	*46*
S_T	S_W	S_B	S_F	s_T	s_W	s_B	s_F
9.4°	9.6°	8.8°	8.7°	11.1°	6.3°	10.8°	10.8°
10.1°		*9.5°*	*9.4°*	*12.6°*		*12.4°*	*12.3°*

Site location, 21.3°N, 157.8°W; 25 lava-flow sites; mean number of samples per core 7.72. Nonitalicized dispersion data are calculated with respect to means. Italicized dispersion data are calculated with respect to a centered axial dipole.

K, k are precision parameters for VGP's and directions, respectively.

S, s are angular standard deviations for VGP's and directions, respectively.

Subscript T indicates total dispersion; W, mean within flow dispersion; B, between-flow dispersion (experimental effects removed); F, ancient VGP to geomagnetic field dispersion (experimental and ancient anomaly effects removed).

nificantly different from the geographic pole if 25 is used as the effective number of independent measurements but is not different if the effective number is reduced to 20. Thus, these data may or may not support Wilson's hypothesis.

The angular variances of the present data have been presented in Table 2 in several different measures so that they can be compared in any form desired with variances previously measured. Because of the fact that the axis of rotation seems to be the main symmetry axis of the geomagnetic field and for other reasons [*Doell*, 1970; *Doell and Cox*, 1971], it is preferred here to use the angular standard deviation of VGP's about the geographic axis as the best measure of paleosecular variation of Brunhes age rocks; that is, the value $S_F = 9.4°$. If *Wilson's* [1970] hypothesis is correct, this measure might not be the best measure. Perhaps one should calculate the variance about the VGP that would be produced from the field of Wilson's dipole in Hawaii. In comparing variances from different areas, the effect would be to produce slightly smaller values everywhere and would not significantly change the relative amounts.

The 95% confidence limits about the value $S_F = 9.4°$ for 25 independent measurements are 7.8°–11.7°; for 20 independent measurements, the range is increased to 7.7°–12.0° [*Cox*, 1969]. The paleosecular variation of the Honolulu volcanic series is thus entirely compatible with the value of 11.0° found from 219 Brunhes age lavas from Hawaii [*Doell and Cox*, 1971]. It can be noted, however, that many of the data in the Island of Hawaii compilation are not statistically independent. This confirmation is further evidence for subdued secular variation and for lack of nondipole geomagnetic-field sources in the central Pacific for extended periods of time.

APPENDIX

The name given each unit is from *Stearns and Vaksvik* [1935], and the page reference, from

21562202, 1972, 11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB077i011p02229 by Curtin University Library, Wiley Online Library on [26/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

the same work, is the place where a description of the unit can be found [see also *Winchell*, 1947]. The quadrangle maps referred to are all 7.5′ series (1:24,000 scale) U.S. Geological Survey topographic maps dated 1959 with the exception of Mokapu, which is dated 1952. Numbers in parentheses are the latitude and longitude, respectively, of the sampling site followed by the maximum lateral distance between individual sample cores (in meters).

HS 27: Sugar Loaf basalt (p. 158). Roadcuts along south side of Lunalilo freeway south of Moiliili Quarry; 200 meters NE of Kuhio School, center Honolulu quadrangle (21.294°N, 202.179°E, 36 meters).

HS 26: Kaupo basalt (p. 154). Just NE of parking lot at Makapuu Beach, east center Koko Head quadrangle (21.317°N, 202.338°E, 60 meters).

HS 25: Kaohikaipu volcanics (p. 153). At sea level on westermost coast of Kaohikaipu Island north of Makapuu Point, east center Koko Head quadrangle (21.322°N, 202.340°E, 20 meters).

HS 24: Kalama volcanics (p. 153). At sea level along Waimanalo Beach Park, 500 meters SW of Kaloko, Makapuu Head, east center Koko Head quadrangle (21.295°N, 202.335°E, 59 meters).

HS 23: Koko volcanics (p. 152). Thin flow in roadcuts on SE side of Highway 72, 650 meters east of Kahauloa crater, SE Koko Head quadrangle (21.281°N, 202.313°E, 29 meters).

HS 22: Punchbowl volcanics (p. 145). Just below top of small ravine on SE side of Punchbowl, center Honolulu quadrangle (21.314°N, 202.157°E, 13 meters).

HS 21: Castle volcanics (p. 145). East and south sides of Kapaa Quarry, west of Kawainui Swamp, SW Mokapu quadrangle (21.398°N, 202.230°E, 190 meters).

HS 20: Black Point basalt (p. 140). West juncture of Kupikipikio Point (Black Point) with Diamond Head coast, 300 meters SE of Kahala avenue and Diamond Head road junction, SE Honolulu quadrangle (21.260°N, 202.213°E, 20 meters).

HS 19: Mauumae volcanics (p. 140). Roadcuts on SE side of Sierra Drive west of Wilhelmina rise summit, 60 meters W-NW of point 334, east center Honolulu quadrangle (21.289°N, 202.199°E, 8 meters).

HS 18: Kaimuki volcanics (p. 138). Thin flow in cone at fire station, south of Pahoa Avenue near 12th avenue, SE Honolulu quadrangle (21.298°N, 202.198°E, 17 meters). Kaimuki volcanics were renamed Kaimuki basalt by *MacDonald and Davis* [1956].

HS 17: Upper basalt of Kaimuki volcanics (p. 138). Roadcuts on south side of Lunalilo freeway between 7th and 8th avenues, east center Honolulu quadrangle. Exposed during construction in 1965 (21.286°N, 202.192°E, 108 meters).

HS 16: Lower basalt of Kaimuki volcanics (p. 138). Roadcuts on south side of Lunalilo Freeway between 6th and 7th avenues, east center Honolulu quadrangle. Exposed during construction in 1965 (21.286°N, 202.191°E, 77 meters).

HS 15: Training School volcanics (p. 132). Roadcuts on north side of highway 6 just east of Maunawili Stream, SW Mokapu quadrangle (21.382°N, 202.239°E, 22 meters).

HS 14: Ainoni volcanics (p. 129). Roadcuts on NW side of unimproved road west of Ainoni Stream, 770 meters south of gaging station on Olomana Stream, NW Koko Head quadrangle (21.357°N, 202.231°E, 27 meters).

HS 13: Kaau volcanics (p. 123). In Pukele Stream at gaging station near Makaoe Spring, east center Honolulu quadrangle (21.930°N, 202.208°E, 7 meters).

HS 12: Pali volcanics (p. 116). Roadcuts on south side of old Pali highway, NE of summit, 320 meters E-SE of BM 649, NE Honolulu quadrangle (21.373°N, 202.212°E, 3 meters).

HS 11: Dike, Pali volcanics (p. 117). Roadcuts on south side of old Pali Highway E-NE of summit, 500 meters SE of BM 649, NE Honolulu quadrangle (21.373°N, 202.208°E, 6 meters).

HS 10: Upper flow of Nuuanu volcanics (p. 112). Roadcuts on NE side of Nuuanu drive, 200 meters NE of gaging station below Alewa Heights Spring, Nuuanu Valley, north center Honolulu quadrangle (21.350°N, 202.174°E, 18 meters).

HS 9: Lower flow of Nuuanu volcanics (p. 112). In Nuuanu Stream under Pali highway bridge just east of Alepena Pool, north center Honolulu quadrangle (21.328°N, 202.152°E, 52 meters).

21562202, 1972, 11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB077i011p02129 by Curtin University Library, Wiley Online Library on [26/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

HS 8: Kaneohe volcanics (p. 111). In Keaahala Stream 50 meters west of easternmost bridge, east center Kaneohe quadrangle (21.418°N, 202.203°E, 8 meters).

HS 7: Kaneohe volcanics (p. 111). In upper reaches of east branch of Kawa Stream, 600 meters south of Kaneohe Bay drive, SE Kaneohe quadrangle (21.404°N, 202.209°E, 12 meters).

HS 6: Haiku volcanics (p. 106). In Haeia Stream, 400 meters W-SW of Baskerville Spring, center Kaneohe quadrangle (21.416°N, 202.179°E, 9 meters).

HS 5: Kalihi volcanics (p. 103). In small draw 400 meters N-NW of Likiliki highway and 750 meters west of the west portal of Wilson tunnel, south center Kaneohe quadrangle (21.363°N, 202.172°E, 22 meters).

HS 4: Rocky Hill volcanics (p. 101). Natural outcrops 60 meters SW of point 296 on top of Rocky Hill, center Honolulu quadrangle (21.306°N, 202.170°E, 6 meters).

HS 3: Hawaiiloa volcanics (p. 99). At sea level on NW side of Pyramid Rock, NW Mokapu peninsula, NW Mokapu quadrangle (21.465°N, 202.233°E, 36 meters).

HS 2: Hawaiiloa volcanics (p. 99). Natural outcrops south of road along north shore of Mokapu peninsula, due north of Puu Hawaiiloa, NW Mokapu quadrangle (21.427°N, 202.242°E, 55 meters).

HS 1: Hawaiiloa volcanics (p. 99). Natural outcrops 500 meters east of tower on summit of Puu Hawaiiloa, NW Mokapu quadrangle (21.462°N, 202.243°E, 6 meters).

Acknowledgments. It is a pleasure to acknowledge the assistance of Edward A. Mankinen, G. Brent Dalrymple, Everett D. Jackson, and Melvin H. Beeson in the location and sampling of these lavas on Oahu. The sloop *Sansara,* her skipper David Smith, and crew are due thanks for transportation to, and landings on, the windward Oahu offshore islands. Permission for these visits was kindly provided by the Director, Division of Fish and Game, Department of Land and Natural Resources, State of Hawaii. Magnetic measurements were made in the laboratory of the U.S. Geological Survey by Edward A. Mankinen. Publication was authorized by the Director, U.S. Geological Survey.

REFERENCES

Broecker, W. S., D. L. Thurber, John Goddard, Teh-Lung Ku, R. K. Matthews, and K. J. Mesolella, Milankovitch hypothesis supported by precise dating of coral reefs and deep-sea sediments, *Science, 159,* 297, 1968.

Cox, Allan, Analysis of present geomagnetic field for comparison with paleomagnetic results, *J. Geomagn. Geoelec., 13,* 101, 1962.

Cox, Allan, Confidence limits for the precision parameter κ, *Geophys. J., 17,* 545, 1969.

Cox, Allan, Latitude dependence of the angular dispersion of the geomagnetic field, *Geophys. J., 20,* 253, 1970.

Cox, Allan, and R. R. Doell, Long period variations of the geomagnetic field, *Bull. Seismol. Soc. Amer., 54,* 2243, 1964.

Creer, K. M., The dispersion of the geomagnetic field due to secular variation and its determination for remote times from paleomagnetic data, *J. Geophys. Res., 67,* 3461, 1962.

Doell, R. R., Paleomagnetism of the Kau volcanic series, Hawaii, *J. Geophys. Res., 74,* 4857, 1969.

Doell, R. R., Paleomagnetic secular variation study of lavas from the Massif Central, France, *Earth Planet. Sci. Lett., 8,* 352, 1970.

Doell, R. R., and Allan Cox, The accuracy of the paleomagnetic method as evaluated from historic Hawaiian lava flows, *J. Geophys. Res., 68,* 1997, 1963.

Doell, R. R., and Allan Cox, Paleomagnetism of Hawaiian lava flows, *J. Geophys. Res., 70,* 3377, 1965.

Doell, R. R., and Allan Cox, Pacific geomagnetic secular variation, *Science, 71,* 248, 1971.

Doell, R. R., and Allan Cox, The Pacific geomagnetic secular variation anomaly and the question of lateral uniformity in the lower mantle, in *The Nature of the Solid Earth,* pp. 245–284, McGraw-Hill, New York, 1972.

Fisher, R. A., Dispersion on a sphere, *Proc. Roy. Soc., London, Ser. A, 217,* 295, 1953.

Fisk, H. W., Isopors and isoporic movement, *Bull. Int. Geod. Geophys. Union,* no. 8, 280, 1931.

Gramlich, J. W., V. A. Lewis, and J. J. Naughton, Potassium-argon dating of Holocene basalts of the Honolulu volcanic series, *Bull. Geol. Soc. Amer., 82,* 1399, 1971.

Macdonald, G. A., and D. A. Davis, Zone des Hawai, in *Océanie,* part 2, vol. 6, pp. 71–143, Centre National de la Recherche Scientifique, 1956.

Ruhe, R. V., J. M. Williams, and E. I. Hill, Shorelines and submarine shelves, Oahu, Hawaii, *J. Geol., 73,* 485, 1965.

Stearns, H. T., Pleistocene shore lines on the Islands of Oahu and Maui, Hawaii, *Bull. Geol. Soc. Amer., 46,* 1932, 1935.

Stearns, H. T., *Geology of the State of Hawaii,* 266 pp., Pacific Books, Palo Alto, Calif., 1966.

Stearns, H. T., and K. N. Vaksvik, Geology and ground-water resources of the Island of Oahu, Hawaii, *Bull. Hawaii Div. Hydrogr., 1,* 478 pp., 1935.

21562202, 1972, 11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB077i011p02129 by Curtin University Library, Wiley Online Library on [26/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Watson, G. S., Analysis of dispersion on a sphere, *Mon. Notic. Roy. Astron. Soc., Geophys. suppl.*, *7*, 153, 1956.

Watson, G. S., and E. Irving, Statistical methods in rock magnetism, *Mon. Notic. Roy. Astron. Soc., Geophys. Suppl.*, *7*, 289, 1957.

Wilson, R. L., Permanent aspects of the earth's non-dipole magnetic field over Upper Tertiary times, *Geophys. J.*, *19*, 417, 1970.

Winchell, Horace, Honolulu series, Oahu, Hawaii, *Bull. Geol. Soc. Amer.*, *58*, 1, 1947.

(Received July 9, 1971;
revised December 30, 1971.)

21562202, 1972, 11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB077i011p02129 by Curtin University Library, Wiley Online Library on [26/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License